

IoT Network Intrusion Detection System

Teammate Setup & Training Manual (Git Collaboration Guide)

PROJECT SCOPE: Computer Networks Semester Project	GITHUB REPOSITORY URL:
BASE PAPER: Wang et al. (Journal of Supercomputing, 2024)	https://github.com/Harish8955/IoT_NIDS_Project.git

STEP 1: Clone Repository to Desired Laptop Destination

Open **Command Prompt (cmd.exe)** on your laptop and navigate to your preferred working folder (Desktop, Downloads, D: drive, etc.):

```
cd /d "C:\Users\YOUR_NAME\Desktop"
git clone https://github.com/Harish8955/IoT_NIDS_Project.git
cd IoT_NIDS_Project
```

STEP 2: Create & Activate Local Virtual Environment (10 Seconds)

Create a fresh local venv bound specifically to your laptop's drive paths:

```
python -m venv venv
venv\Scripts\activate.bat
```

STEP 3: Install Dependencies (Single Command)

Install PyTorch, CTGAN, Scikit-Learn, Pandas, ReportLab, and all 13 required libraries:

```
pip install -r requirements.txt
```

Note for NVIDIA GPUs: Install CUDA PyTorch via `pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118` for 10x faster training.

STEP 4: Pre-Combined Master Datasets (Ready to Train)

The master datasets are **already pre-combined** inside the shared Google Drive / project package (`data/`). Teammates can skip dataset combining and proceed directly to model training!

- data/UNSW_NB15_combined.csv (Pre-combined UNSW-NB15 dataset)
- data/CIC_IDS2018_combined.csv (Pre-combined CSE-CIC-IDS2018 dataset)
- data/CICIOT23_combined.csv (Pre-combined CIC-IOT2023 dataset)

STEP 5: Execute Assigned Model Training Commands

Task Category	Command Prompt Execution String
Proposed Model Peak Runs	<code>python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model proposed --epochs 200 --balance</code>
Baseline 1D-CNN Run	<code>python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model cnn --epochs 50 --balance</code>
Baseline ResNet Run	<code>python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model resnet --epochs 50 --balance</code>
Dual-Engine LOCO Zero-Day	<code>python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model dual-engine --epochs 50 --balance --hide_class Worms</code>

STEP 6: Push Results Back to Master GitHub Repository

When training completes, sync your generated metrics back to the team's master Excel summary sheet:

```
git pull
git add results/
git commit -m "Added experiment training results"
git push
```